3.5 Completing the Square

Name _____ Period ____ Date _____

Algebra II 3.5 (30-minute plan) · Student sheet

I can…

- solve equations using square roots when one side is a perfect square.
- complete the square and solve quadratic equations.
- complete the square to write quadratic functions in vertex form.

Vocabulary · fill in the missing word

Completing the square — _____ $(\frac{b}{2})^2$ to $x^2 + bx$ to make a perfect square trinomial: $x^2 + bx + (\frac{b}{2})^2 = (x + \frac{b}{2})^2$

Part 1 · Notes and examples (fill in each note, then work the example)

Completing the square: To complete the square for $x^2 + bx$, add _____.

Example 1

Find the value of c that makes $x^2 - 14x + c$ a perfect square trinomial. Write it as a square.

Solving: When solving, add it to _____ sides.

Example 2

Solve $x^2 - 4x - 3 = 0$ by completing the square.

When a ≠ 1: If $a \neq 1$, first _____.

Example 3

Solve $2x^2 + 8x + 10 = 0$ by completing the square.

Part 2 · Partner practice

Partner practice

Solve $x^2 + 8x + 1 = 0$ by completing the square.